

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE CODE: CSA14

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO1: Apply lexical analysis for token specification and recognition using LEX tool. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

**Duration :60 Mins
On Class
Total Marks:50**

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

*I **_192511145_ sheik Mohamed m**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Sheik Mohamed.m

$$1) x * y - z + 20 = w$$

Analytical Problem Solving

$$w = x * y - z + 20$$



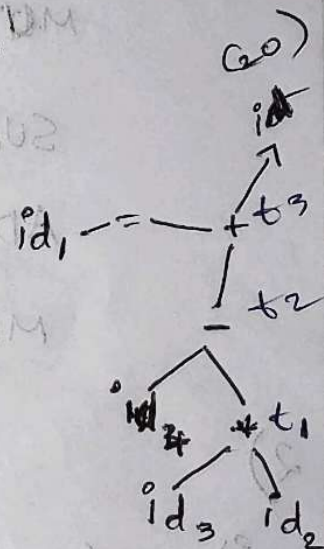
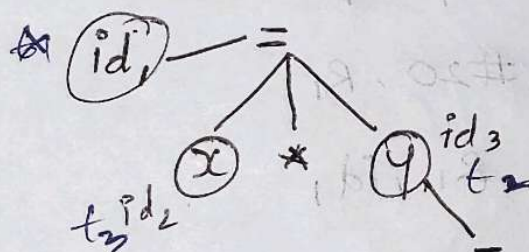
lexical Analyzer

$$\langle id_1, w \rangle \Rightarrow \langle id_2, x \rangle \langle * \rangle \langle id_3, y \rangle$$

$$\rightarrow \langle id_4, z \rangle \langle + \rangle \langle 20 \rangle$$



semantic analyzer



Parse tree

(20)

Intermediate code generator

$$t_1 = id_3 * id_2$$

$$t_2 = id_4 - t_1 \rightarrow t_1 - id_4$$

$$t_3 = t_2 + 20$$

$$id_1 = t_3$$

Code optimization generator

$$t_1 = id_2 * id_3$$

$$id_1 = (t_1 - id_4) + 20.$$

Code generator

$$R_1 \leftarrow id_2$$

$$R_2 \leftarrow id_3$$

MOV id_2, R_1

MOV id_3, R_2

MUL R_2, R_1

SUB id_4, R_1

ADD $\#20, R_1$

MOV R_1, id_1

2)

$$i) 1(0+1)^*1+1$$

$$ii) (0+1)^*01(0+1)^*$$

$$iii) (1^*01^*)$$

$$iv) (0+1)^*1(0+1)^*1(0+1)^*$$

$$v) (0+1)^*1+1$$

3)

i) $(\epsilon + aa^*)^*$

ii) $(a^* + b^*)^* abb$

iii) $(a^* b^*)^* aba$

iv) $(a + ab)^*$

v) $(abc)^* + (a^* b^*)^*$

i) $(a^*)^* = a^*$

$L = \{ \epsilon, a, aa, aaa, \dots \}$

ii) $(a^* b^*)^* = (a^* + b^*)$

$= (a + b)^*$

$= (a + b)^* aba$

$L = aba$

iii) $(a^* b^*)^* = (a + b)^*$

$= (a + b)^* aba$

$L = aba$

iv) $(a^* + b^*)^* abb$

$(a^* + b^*)^* abb = (a + b)^* abb$

$L = abb$

v) $(a + ab)^*$

$L = \{ \epsilon, a, aa, ab, aab, aba, \dots \}$

vi) $(aba)^* + (a^* b^*)^*$

$$(a^*b^*)^* = (a+b)^*$$

$$(aba)^* = \{ \epsilon, aba, ababa, \dots \}$$

4)

$$1) [a-zA-Z][a-zA-Z0-9]^*$$

program:

```
%{
```

```
#include <stdio.h>
```

```
%}
```

```
letter [A-Z a-z]
```

```
digit [0-9]
```

```
identifier {letter} ({letter} {digit})^*
```

```
% %
```

```
if { printf ("xx text"); }
```

```
else { printf ("yy text"); }
```

```
while { printf ("yy text"); }
```

```
{ id } { printf ("yy text"); }
```

```
[0-9]+ { printf ("xx text"); }
```

```
" + "/" - "/" { printf ("xx text"); }
```

```
(")",) / { id } { printf ("yy text"); }
```


{ 1 + 1n } + ; (1+0) = * (1+0) (1)

print (xxtext) ; }

int main () {

print ("Enter")

xxlex ()

return 0 ; }

int xxwarp () {

return ;

}

ii)

* The lexical analyzer will consume longest possible string that matches a pattern

numeric formats : (*x)

* Distinguish between valid &

invalid number formats.

* Let, 2 pattern match the

some input string.

$$5) i) (0^* 1^*)^* = (0+1)^*$$

$$(0^* 1^*)^* = \{ \epsilon, 0, 1, 00, 01, 11, 10, \dots \}$$

$$(0+1)^* = \{ \epsilon, 0, 1, 00, 01, 11, 10, \dots \}$$

$$(0^* 1^*)^* = (0+1)^*$$

$$ii) (0+1)^* (0+1)^* = (0+1)^*$$

$$R^* R^* = R^*$$

$$R^* R^* = R^*$$

$$iii) (0^*)^* = 0^*$$

$$(R^*)^* = R^*$$

$$IV) (0+1)^* 0 (0+1)^* + (0+1)^* 1 (0+1)^* = (0+1)^*$$

$$(0+1)^* 0 (0+1)^* + (0+1)^* 1 (0+1)^* = (0+1)^*$$

$$(0+1)^* (1+0) = (0+1)^*$$

$$(0+1)^* = (0+1)^*$$

$$V) \epsilon + 0(0)^* = 0$$

$$\epsilon + R(R^*) = R^*$$

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation